

E-Commerce Data Quality Assessment and Business Report

Objective

The objective of this project was to identify data quality issues, clean the data using Pandas, and compare business reporting before and after cleaning.

Business Problem

E-commerce businesses rely on accurate data for:

- Revenue Reporting
- Customer Analysis
- Product Performance Tracking
- Operational Monitoring

However, poor data quality can result in:

- Inflated Revenue
- Incorrect Order Counts
- Misleading Regional Analysis
- Inaccurate Customer Insights

Dataset Overview

Dataset Characteristics:

- 230,000 records
- Customer transactions
- Product information
- Payment details
- Order status information
- Customer ratings

Data Quality Issues Identified

The following issues were identified:

- Duplicate transactions
- Missing values
- Inconsistent Categories
- Gender Inconsistency
- Invalid Values

Data Cleaning Process

- `drop_duplicates()`
- `fillna()`
- `isnull()`
- `groupby()`
- `replace()`
- `str.strip()`
- `astype()`
- `to_datetime()`
- `describe()`
- `info()`

Business Impact of Data Cleaning

1. Revenue Accuracy:

Before Cleaning:

- ₹1,980.75M

After Cleaning:

- ₹1,925.17M

Impact:

Revenue reduced by ₹55.58M

This indicates inflated revenue caused by duplicate or invalid transactions.

2.Order Count Accuracy**Before Cleaning:**

- 230.00K

After Cleaning:

- 215.05K

Impact:

14.95K records removed

This improved reporting accuracy.

3.Customer Segmentation

Gender values were standardized from multiple inconsistent formats into two business-ready categories.

Result:

- Cleaner customer analysis
- Improved reporting consistency

4.Category Reporting

Product categories were standardized and consolidated.

Result:

- Accurate category-level revenue analysis
- Better product performance evaluation

Key Business Insights:

- Home and Electronics generated the highest revenue.
- Credit Card and Wallet transactions generated the largest share of revenue.
- Running Shoes and Wireless Mouse were among the top-selling products.
- Cancellation and Return Rates remained close to 20%, indicating opportunities for operational improvement.
- Revenue distribution across genders was nearly balanced.

Tools Used

- Python
- Pandas
- NumPy
- Jupyter Notebook
- Tableau

Conclusion

The project demonstrated how poor data quality can significantly affect business reporting. Through systematic data cleaning and validation using Pandas, inaccurate records were corrected, resulting in more reliable KPIs and business insights. The comparison between raw and cleaned dashboards highlights the importance of data quality in decision-making and reporting.

Dashboards

